

TOPIC 6.A

Our immune system and health

1. Relate the concept of health to antigens and the functioning of the immune system!
2. Introduce the characteristic cells of the immune system and the essence of their functioning!
3. Compare the processes of active and passive, natural and artificial immunisation!
4. Introduce Ignác Semmelweis's research method, results, and their significance!
5. Interpret the differences in the treatment of diseases of viral and bacterial origin!

Expected knowledge	Attainable score	Attained score
1. • The healthy body resists infections / maintains its self-identity. • In this, it is aided by the functioning of the immune system, which acts against antigens, • i.e., against molecules that are chemically foreign to the body / the organisms carrying them.	3 points	
2. • Phagocytes destroy pathogens by engulfing them (phagocytosis) / they present the antigens to other members of the immune system. • The mass of dead phagocytes and pathogens forms pus. • Lymphocytes act by destroying pathogenic cells / infected self-cells or through the production of antibodies / immunoglobulins.	5 points	
3. • [Active immunisation stimulates the body's own immune system to produce antibodies.] • During passive immunisation, antibodies (produced by another organism) are directly introduced into the body. • Natural immunisation always develops under natural conditions, e.g., due to antibodies	5 points	

passing from the mother's body into the offspring. • The essence of artificial immunisation is the immunising agent introduced via vaccination.		
4. • By recognising the cause of puerperal fever (childbed fever) • and by eliminating the infection / blocking its transmission, Semmelweis became important in the history of medicine. • The method of recognition / proof: statistical analysis (mortality data). • His procedure: disinfection / hygiene.	4 points	
5. • Antibiotics can be effective against bacteria, • but not against viruses, • because viruses do not have their own (independent) metabolism.	3 points	
6. Structure of the answer, communication skills	5 points	
Total:	30 points	

TOPIC 6.B

Comparison of two animal species

1. You will find two animal species in the picture.
2. Name and compare them based on the criteria given on the worksheet.



Worksheet - animal identification

Animal 1	Comparison criteria	[Animal 2]
	Name	
	Phylum, class	
	Integument (Skin)	
	Respiration	
	Reproduction	
	Development	
...	...	...

Expected knowledge	Attainable score	Attained score
The identification of the two species and comparison of the two species(white stork, fire salamander)	20 points	